

PhaseAlign: Complex Phase Alignment for Stable Open-Vocabulary Semantic Segmentation

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Abstract

Open-Vocabulary Segmentation (OVS) aims to achieve pixel-level semantic recognition from arbitrary text queries. Existing large-scale visual-linguistic models, such as CLIP, perform well in zero-shot generalization, but their image-level training objectives and real-valued cross-modal alignment mix amplitude and phase information, limiting fine-grained segmentation and often causing blurred boundaries and fragmented structures. Inspired by the ability of electromagnetic wave phase to control interference independently of amplitude, we propose PhaseAlign, an OVS framework based on Complex Phase Alignment (CPA). CPA explicitly decouples the magnitude and phase of visual and textual embeddings in the complex domain, refining effective features for stable cross-modal alignment. To further enhance structural awareness, we introduce spatial-aware cross-modal projection, which models local neighborhood relations via multi-scale spatial contrast normalization, and attention-guided affinity modeling, which leverages pretrained ViT self-attention to propagate category activations, improving boundary clarity and region integrity. Experiments show that PhaseAlign achieves state-of-the-art performance on multiple OVS benchmarks.

1. Introduction

Semantic segmentation is a core task of pixel-level visual understanding, with critical applications in autonomous driving, remote sensing, and scene understanding. However,

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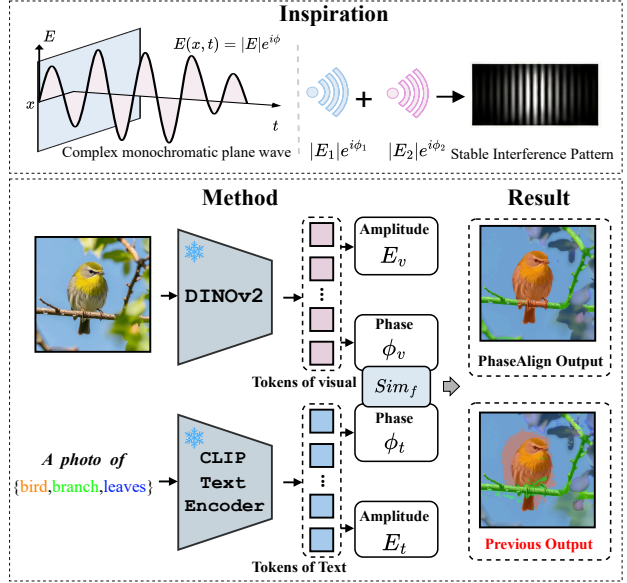


Figure 1. Inspired by phase-governed electromagnetic interference, PhaseAlign decouples amplitude and phase in the complex domain to enable stable pixel-level cross-modal alignment for open-vocabulary segmentation.

traditional semantic segmentation (Ronneberger et al., 2015; Cheng et al., 2022) methods usually rely on closed category sets for training, an assumption that is difficult to hold in open environments, causing the models to exhibit significant generalization bottlenecks when facing unseen categories. To break through this limitation, Open-Vocabulary Segmentation (OVS) (Zhao et al., 2017) aims to achieve pixel-level segmentation of unknown categories given only images with arbitrary text queries, thus providing higher flexibility and scalability for visual understanding in open scenes.

Recent advances in large-scale vision-language pretraining (Cherti et al., 2023), exemplified by CLIP (Radford et al., 2021) and ALIGN (Jia et al., 2021), map visual and textual representations into a shared semantic embedding space through contrastive learning on image-text pairs. These models demonstrate remarkable generalization in zero-shot classification and cross-modal retrieval, providing a natural foundation for OVS. However, the training objectives of these models focus primarily on image-level representations, and their cross-modal consistency is largely global rather

than at the pixel level. Directly transferring such models to dense prediction tasks often produces blurred boundaries and fragmented local structures, limiting their applicability for fine-grained segmentation.

Further analysis suggests that the performance bottleneck of existing OVS methods is not solely due to insufficient supervision, but is also closely related to limitations of the prevailing cross-modal alignment paradigm (Kang et al., 2025; Han et al., 2025). Most approaches rely on magnitude-sensitive similarity metrics, such as cosine similarity or inner product, which implicitly assume that visual and textual embeddings are comparable in scale and that semantic information is jointly encoded in both magnitude and direction. However, visual and textual encoders exhibit fundamentally different modeling characteristics, leading to heterogeneous magnitude distributions and modality-specific noise. When such real-valued similarity measures are directly extended to pixel-level alignment, scale perturbations can significantly distort the optimization signal, resulting in unstable feature refinement and fragmented local structures. Consequently, these metrics often struggle to provide sufficiently clear, stable, and structurally meaningful supervision for fine-grained cross-modal alignment.

These observations suggest that high-quality cross-modal alignment should not only measure semantic consistency but also provide structurally clear and gradient-stable constraints for feature refinement. **Inspired by the complex-valued representation of electromagnetic waves, where in a monochromatic plane wave $E(x, t) = |E|e^{i\phi}$ the amplitude $|E|$ primarily governs energy distribution while the phase ϕ encodes propagation state, interference experiments further demonstrate that as long as the phase remains consistent, stable interference patterns can form even if amplitudes differ significantly (Born & Wolf, 1999).** Based on this insight, we propose Complex Phase Alignment (CPA), which maps visual and textual embeddings into the complex domain, explicitly disentangles magnitude and phase in the representation, and uses phase differences—rather than amplitude correlation—as the cross-modal alignment signal. This design helps mitigate the impact of magnitude bias on optimization and provides a more stable and discriminative signal for pixel-level feature refinement.

However, accurate pixel-level prediction requires more than a robust alignment metric. CPA operates on local token-level representations, whose effectiveness depends on structured spatial features. Conventional token-wise MLP projections ignore the spatial organization of visual features, limiting the capacity of phase alignment. To address this, we introduce a Spatial-Aware Cross-modal Projection (SACP) module, which explicitly models local neighborhood relations through multi-scale spatial normalization, enhancing

spatial coherence and providing a structured representation for phase alignment.

Even with structured cross-modal alignment, translating semantic consistency into precise segmentation boundaries remains challenging. Vision Transformer self-attention implicitly captures long-range semantic affinities during pre-training, but such information is underutilized when generating class activation maps. To leverage this, we design an attention-guided affinity modeling (AAM) module, which constructs semantic affinity graphs from pretrained self-attention and performs constrained diffusion to propagate class activation responses within semantically consistent regions. This refinement enforces structural coherence, sharpens boundaries, and improves region completeness. In summary, this work presents a theory-driven framework for open-vocabulary segmentation. Our main contributions are as follows:

- (1) We present PhaseAlign, a theory-driven OVS framework that leverages complex phase alignment to achieve stable pixel-level visual–language correspondence beyond magnitude-based similarity.
- (2) We introduce a spatial-aware cross-modal projection, which explicitly models local spatial structure via multi-scale normalization to provide structured representations for phase alignment.
- (3) We develop an attention-guided affinity modeling module, leveraging ViT self-attention to propagate class activation responses and enhance fine-grained structural and boundary quality.
- (4) Extensive experiments on multiple OVS benchmarks validate the effectiveness of PhaseAlign and demonstrate the complementary benefits of its individual components.

2. Related work

2.1. Open-Vocabulary Segmentation

Open-Vocabulary Segmentation (OVS) aims to enable pixel-level prediction conditioned on arbitrary textual queries during inference (Mukhoti et al., 2023; Rao et al., 2022; Bucher et al., 2019), overcoming the limitations of traditional semantic segmentation that rely on a fixed set of categories. Early approaches mitigated the generalization gap for unseen classes by leveraging external class mappings, hierarchical semantics, or additional annotated data. However, these methods remain constrained in terms of category scalability and inference flexibility. With the advent of large-scale vision-language pretraining, cross-modal alignment based on models such as CLIP (Radford et al., 2021) has become the dominant paradigm for OVS. These approaches typically project image features and textual embeddings into a shared semantic space (Liu et al., 2025), generating class

activation maps (Dong et al., 2023) or pseudo labels through similarity computation, which are subsequently used for segmentation. While this strategy improves scalability in open environments without additional category supervision, existing OVS methods largely inherit the modeling assumptions of vision-language models in image-level tasks. Their cross-modal consistency is primarily global, and explicit modeling of pixel-level structures and local semantic relationships remains insufficient.

2.2. Cross-Modal Alignment and Pixel-Level Projection

Cross-modal alignment constitutes the core challenge in open-vocabulary segmentation (OVS). Most existing methods (Radford et al., 2021; Wysoczańska et al., 2024; Lan et al., 2024) rely on cosine similarity or inner product to measure visual-textual correspondence, extending image-level alignment metrics to pixel or token representations. While effective in practice (Barsellotti et al., 2024; Liu et al., 2024), this paradigm implicitly assumes that visual and textual embeddings are comparable in magnitude, an assumption that often breaks down at the pixel level (Rao et al., 2022; Dong et al., 2023; Jose et al., 2025). Differences in statistical distributions, normalization schemes, and semantic encoding mechanisms across modalities render magnitude-sensitive metrics susceptible to noise and scale mismatch, leading to unstable alignment in fine-grained scenarios. To enable dense prediction, some approaches introduce token-wise projection heads (Xu et al., 2022; Fu et al., 2024) or channel-level adaptation modules (Xu et al., 2023). However, these projections typically disregard the spatial organization of visual features (Cho et al., 2024; Jing et al., 2024), treating tokens as independent units and failing to capture stable local contrast. As a result, the lack of explicit spatial modeling further exacerbates alignment uncertainty and limits pixel-level semantic consistency in OVS.

2.3. Attention-Guided Structural Consistency

Improving spatial consistency and boundary quality in semantic segmentation has motivated a range of refinement strategies, including feature similarity-based region propagation, diffusion mechanisms (Wang et al., 2025a), and conditional random fields (Wang et al., 2023). More recently, Vision Transformer self-attention has been shown to implicitly capture long-range semantic affinities between pixels (Radford et al., 2021; Yang et al., 2024; Lan et al., 2024), and has been leveraged to guide structural optimization in open-vocabulary segmentation (OVS). Several methods utilize attention maps (Huang et al., 2022; Dong et al., 2023) or feature affinity matrices to smooth or diffuse class activation maps, reducing prediction noise and enhancing regional coherence. However, existing approaches typically treat attention as a generic smoothing prior, without explicitly distinguishing semantically consistent regions from

those prone to erroneous propagation. Moreover, the diffusion process is often weakly coupled with cross-modal alignment, which may lead to semantic drift or boundary shifts despite improved structural continuity. Effectively exploiting attention-induced structural information under explicit cross-modal semantic constraints therefore remains an open challenge in OVS.

3. Methodology

3.1. Preliminaries

Open-Vocabulary Segmentation (OVS) aims to perform pixel-level semantic prediction conditioned on arbitrary textual queries, relaxing the closed-set assumption of conventional semantic segmentation. Given an input image I and a set of text queries $T = \{t_1, \dots, t_N\}$, OVS predicts a pixel-wise mapping $M : \Omega \rightarrow T \cup \{\emptyset\}$, where Ω denotes the image domain and $\emptyset$ represents background. Unlike traditional methods defined over a fixed category set C , OVS relies on learning a shared semantic space Z that aligns visual features $f_v(I)$ and textual embeddings $f_t(t)$, enabling flexible “any-class, on-demand” inference. Large-scale vision-language models such as CLIP (Radford et al., 2021) provide the foundation for OVS by jointly training an image encoder f_v and a text encoder f_t via contrastive learning. Both modalities are projected into a normalized embedding space, and cross-modal similarity is measured using temperature-scaled cosine similarity, $s_{ij} = f_v(I_i)^\top f_t(t_j) / \tau$. The resulting global image-text alignment facilitates generalization to unseen categories and serves as the basis for dense cross-modal correspondence in OVS.

3.2. Complex Phase Alignment (CPA)

A central challenge in cross-modal alignment is the feature heterogeneity between visual and textual encoders: visual representations emphasize local structures and continuous statistical patterns, while textual representations rely on discrete semantic compositions and contextual dependencies. Directly using magnitude-sensitive metrics such as cosine similarity or inner product for pixel-level token alignment is prone to scale noise, which can lead to unstable optimization.

To address this, we propose Complex Phase Alignment (CPA), which maps visual features v and textual features t into a complex-valued space and uses the phase difference, rather than amplitude correlation, as the core measure of semantic consistency. Concretely, given visual and textual features $v \in \mathbb{R}^C$ and $t \in \mathbb{R}^C$ projected into a shared embedding space, we first split the channel dimension into real and imaginary parts to construct complex representations:

$$Z_V = v_r + iv_i, \quad Z_T = t_r + it_i, \quad (1)$$

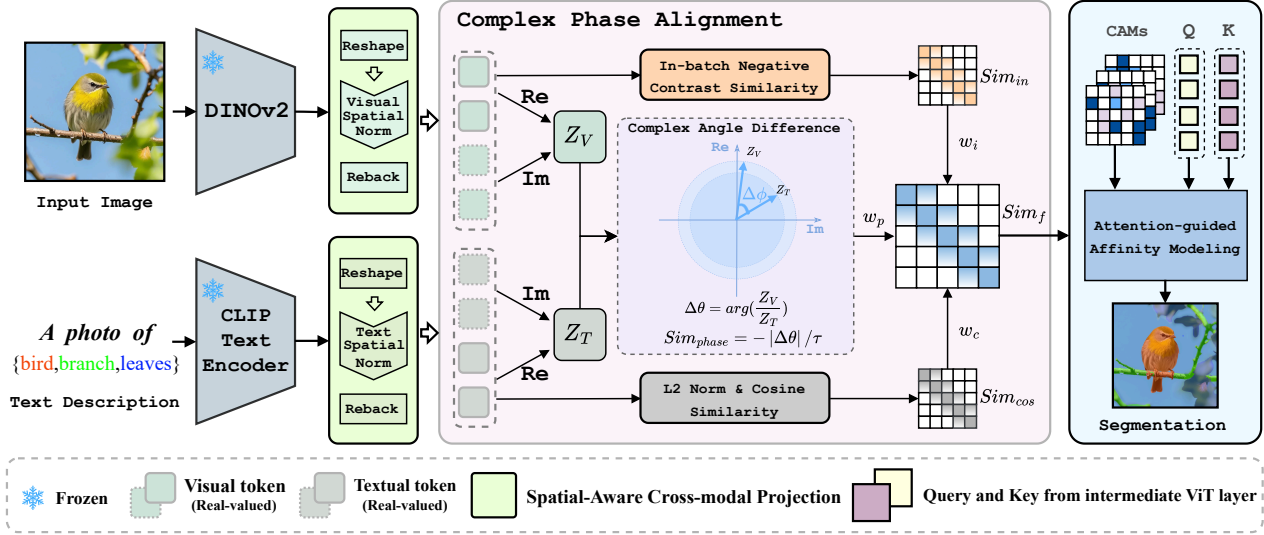


Figure 2. Overview of PhaseAlign. PhaseAlign maps CLIP textual embeddings and DINOv2 visual features into a shared space via Spatial-Aware Cross-modal Projection (SACP), introducing local spatial contrast. In the complex domain, phase-centric similarity ensures stable pixel-level alignment. Class activation maps (CAMs) are computed from the hybrid similarity score S_f , combining phase, cosine, and contrastive terms. Attention-guided Affinity Modeling (AAM) refines spatial structures using CAMs and ViT attention from intermediate layers.

where $v_r, v_i, t_r, t_i \in \mathbb{R}^{C/2}$ denote the respective real and imaginary components along the channel dimension. The relative semantic transformation between the two modalities is explicitly modeled via complex division: $Z_{V \rightarrow T} = Z_V / (Z_T + \epsilon)$, and the phase of this transformation serves as the key cross-modal alignment metric:

$$\Delta\phi = \arg(Z_{V \rightarrow T}) = \arg(Z_V) - \arg(Z_T). \quad (2)$$

This formulation converts cross-modal similarity from magnitude-dependent inner product matching to a measure of relative consistency in semantic direction. The phase-based similarity is then defined as the negative scaled absolute phase difference:

$$\text{Sim}_{\text{phase}}(v, t) = -\frac{1}{\tau} \|\Delta\phi\|. \quad (3)$$

To further improve optimization stability and feature discriminability, we introduce two auxiliary regularization constraints in addition to the phase-based alignment signal. (1) Global direction consistency, where the normalized cosine similarity S_{cos} is retained as a regularization term to preserve coarse directional alignment in the shared embedding space; (2) In-batch contrastive regularization, denoted as S_{in} , which explicitly enhances inter-sample discriminability by leveraging non-matching pairs within each mini-batch. Importantly, complex phase alignment serves as the primary cross-modal supervision signal for pixel-level feature refinement, while the above two terms act as lightweight regularizers to stabilize optimization and prevent representation degeneration. The final similarity score is formulated

as a weighted combination of the three components:

$$S_f = \frac{w_p S_{\text{phase}} + w_c S_{\text{cos}} + w_i S_{\text{in}}}{w_p + w_c + w_i}, \quad (4)$$

where w_p, w_c , and w_i control the relative contributions of phase alignment, global cosine consistency, and batch-wise contrastive regularization, respectively. Unless otherwise specified, we adopt $[w_p, w_c, w_i] = [1, 1, 1]$, which is empirically found to provide a stable and effective balance across datasets. Notably, performance is relatively insensitive to moderate variations of these weights, and the phase term provides the primary directional signal for optimization, while auxiliary terms mainly stabilize convergence.

3.3. Spatial-Aware Cross-Modal Projection

The effectiveness of Complex Phase Alignment (CPA) benefits from structured representations of local tokens. If each token is mapped independently without spatial contrast, the directional consistency in CPA becomes unstable and increasingly noisy. To address this, we propose Spatial-Aware Cross-Modal Projection (SACP), which explicitly models the spatial structure and local neighborhood relations of visual features prior to cross-modal alignment.

Given a visual feature map $X \in \mathbb{R}^{B \times C \times H \times W}$, we introduce a multi-scale spatial contrast normalization operator $\mathcal{N}_k^{(s)}$ in the visual branch to explicitly capture local relative contrasts and enhance spatial structure awareness. For the input feature X , local contrast responses are computed at multiple spatial scales $\{k_s\}_{s=1}^S$:

$$X^{(s)} = \mathcal{N}_k^{(s)}(X), \quad s = 1, \dots, S. \quad (5)$$

where $\mathcal{N}_k^{(s)}$ performs contrast normalization over a local neighborhood of size k_s , suppressing absolute magnitude variations and emphasizing relative changes to highlight structural boundaries and local semantic differences.

A scale attention mechanism is then applied to adaptively fuse features across spatial scales:

$$X_{\text{ms}} = \sum_{s=1}^S \text{Softmax}_s(\psi(\text{GAP}(X))) \odot X^{(s)}. \quad (6)$$

where $\text{GAP}(\cdot)$ denotes global average pooling over spatial dimensions to extract channel-wise context, $\psi(\cdot)$ is a lightweight scale attention mapping composed of two pointwise (1×1) convolutions and non-linear activations to produce multi-scale channel attention, and $\text{Softmax}_s(\cdot)$ normalizes across the scale dimension s to obtain adaptive weights for each channel. The final spatial-aware projection feature is obtained in residual form:

$$\tilde{X} = X + \alpha X_{\text{ms}} + \beta, \quad (7)$$

where α and β are learnable affine parameters to stabilize training.

For the textual branch, since token sequences do not possess explicit two-dimensional spatial topology, we do not apply spatial contrast normalization directly. Given a textual embedding $t \in \mathbb{R}^{B \times C}$ from CLIP, we first reshape it to a spatial form $T \in \mathbb{R}^{B \times C \times 1 \times 1}$, treating the text feature as a degenerate single-point spatial map. A convolutional projection operator $P_c(\cdot)$, consisting of a 1×1 convolution and optionally non-linear and hidden convolution layers, is applied to perform channel reorganization and cross-modal dimension alignment. This projection shares the same operator paradigm as the visual branch, ensuring that visual and textual features reside in a consistent *spatialized representation space* prior to alignment.

3.4. Attention-guided Affinity Modeling

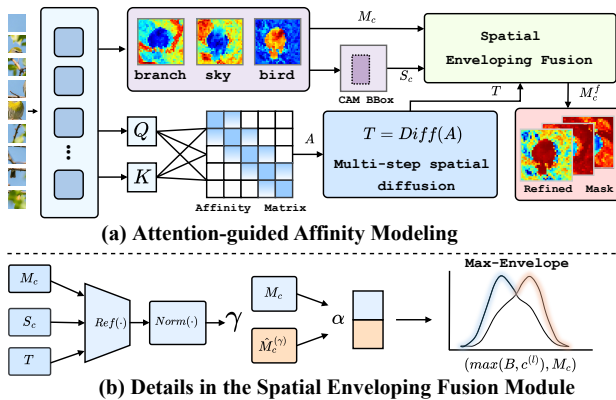


Figure 3. Overview of the Attention-guided Affinity Modeling module for structure-aware refinement.

To enhance the spatial structure and boundary fidelity of segmentation results, we propose an Attention-guided Affinity Modeling (AAM) module, which exploits long-range pixel dependencies encoded in Vision Transformer (ViT) self-attention to perform structure-aware refinement of class activation maps (CAMs), enabling fine-grained segmentation refinement. Given a class-specific CAM $M_c \in \mathbb{R}^{H \times W}$, derived from the text-conditioned similarity map (SimMap), together with the query-key representations Q, K extracted from an intermediate ViT layer, we construct a semantic affinity matrix over spatial tokens based on multi-head self-attention. Specifically, the affinity matrix is computed by scaled dot-product attention and averaged across heads:

$$\mathbf{A} = \frac{1}{H} \sum_{h=1}^H \text{Softmax} \left(\frac{Q_h K_h^\top}{\sqrt{d}} \right). \quad (8)$$

After removing the global CLS token, $\mathbf{A} \in \mathbb{R}^{HW \times HW}$ explicitly captures pixel-level long-range dependencies, encoding intra-object consistency and inter-region semantic connectivity.

To obtain a stable and symmetric propagation structure, we apply a diffusion operator consisting of row normalization, column normalization, and symmetrization, followed by a single self-multiplication to enhance higher-order transition relations. The spatial transition matrix is defined as

$$\mathbf{T} = \left(\frac{1}{2} (\mathcal{N}_c(\mathcal{N}_r(\mathbf{A})) + \mathcal{N}_c(\mathcal{N}_r(\mathbf{A}))^\top) \right)^2, \quad (9)$$

where $\mathcal{N}_r(\cdot)$ and $\mathcal{N}_c(\cdot)$ denote row-wise and column-wise normalization operators, respectively. A class-specific affinity mask $\mathbf{S}_c \in \{0, 1\}^{HW}$ is derived from high-confidence regions of the original CAM to constrain diffusion within potential target regions, preventing propagation across semantically unrelated areas. Under this constrained diffusion formulation, the refined response for each class is computed as

$$\tilde{M}_c = \text{Ref}((\mathbf{T} \odot \mathbf{S}_c) \text{vec}(M_c)), \quad (10)$$

where $\odot$ denotes element-wise masking and $\text{Ref}(\cdot)$ reshapes the vectorized output back to the spatial layout. This refinement propagates high-confidence activations along semantic-consistency paths learned by the ViT, filling internal holes, enhancing local coherence, and recovering more complete object shapes. To further sharpen structural boundaries while ensuring segmentation coverage does not degrade, we apply class-wise normalization and γ -correction, followed by linear fusion with the original CAM and a pixel-wise upper-envelope operation:

$$\widehat{M}_c = \frac{\tilde{M}_c - \min(\tilde{M}_c)}{\max(\tilde{M}_c) - \min(\tilde{M}_c) + \epsilon}, \quad (11)$$

$$\widehat{M}_c^{(\gamma)} = \text{clamp}(\widehat{M}_c, 0, 1)^\gamma, \quad \gamma > 0, \quad (12)$$

$$M_c^f = \max \left((1 - \alpha)M_c + \alpha \widehat{M}_c^{(\gamma)}, M_c \right). \quad (13)$$

Here, $\text{clamp}(\cdot)$ denotes element-wise truncation, and γ amplifies high-confidence regions while suppressing low-response noise. This design substantially improves boundary clarity and regional consistency without compromising the original spatial coverage.

3.5. Training Procedure

We emphasize that PhaseAlign does not alter the contrastive learning objective itself; instead, it reshapes the cross-modal similarity geometry within the InfoNCE framework via complex phase alignment. To optimize cross-modal alignment, we adopt a symmetric InfoNCE objective for contrastive learning over image-text pairs within each mini-batch. Given a batch of B pairs $\{(I_i, T_i)\}_{i=1}^B$, we compute similarities between the spatially projected visual embeddings v_i and textual embeddings t_i , treating matched pairs as positives and all others as negatives. The resulting loss is formulated as

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(S_f(v_i, t_i))}{\sum_{j=1}^B \exp(S_f(v_j, t_i))} + \log \frac{\exp(S_f(v_i, t_i))}{\sum_{j=1}^B \exp(S_f(v_i, t_j))} \right]. \quad (14)$$

Here, $S_f(\cdot, \cdot)$ denotes the proposed hybrid similarity (Eq. 4), which explicitly integrates complex phase alignment to emphasize directional semantic consistency while reducing sensitivity to magnitude variations.

During training, all CLIP backbone parameters are frozen to preserve pretrained open-vocabulary knowledge and prevent overfitting to dataset-specific biases. Only the projection and refinement modules are optimized using AdamW with a polynomial learning rate schedule, enabling stable convergence and efficient learning of fine-grained pixel-level cross-modal alignment.

4. Experiments

4.1. Datasets

To comprehensively evaluate the effectiveness and robustness of the proposed method, we conduct extensive experiments on multiple representative semantic segmentation benchmarks, including PASCAL VOC (Everingham et al., 2012), PASCAL Context (Mottaghi et al., 2014), and COCO-Stuff (Caesar et al., 2018). Following prior OVS evaluation protocols, these datasets are further organized into two experimental configurations according to category definitions: settings with a background class (V20, PC59, and Stuff) and settings without a background class (V21,

PC60, and Object). In addition, we report results on the more challenging ADE20K(ADE) (Zhou et al., 2017) and Cityscapes(City) (Cordts et al., 2016) benchmarks to assess generalization across diverse scenes and data distributions.

4.2. Implementation Details

In the main experimental setup, we adopt DINOv2 ViT-B/14 and DINOv2 ViT-L/14 as the visual backbones for the base and large model configurations, respectively. Both variants are paired with a CLIP ViT-B/16 (Dosovitskiy, 2020) text encoder. All experiments employ the DINOv2 model with register tokens (Darcet et al., 2024). Training is performed using the Adam optimizer with a batch size of 128, a fixed learning rate of 1×10^{-4} , and a total of 100 epochs. The training data are drawn from the COCO Captions 2014 training split, which contains approximately 80K images.

4.3. Comparison with Previous Methods

We systematically compare the proposed method with a diverse set of representative unsupervised open-vocabulary semantic segmentation approaches, with quantitative results summarized in Table 1. The compared methods include early CLIP-based baselines (e.g., CLIP (Radford et al., 2021), GroupViT (Xu et al., 2022)), contrastive learning-based approaches (e.g., TCL (Cha et al., 2023), SegCLIP (Wang et al., 2024)), and recent methods that incorporate external visual priors or reference construction mechanisms, such as CoDe (Wu et al., 2024), CLIP-DINOiser (Wysoczańska et al., 2024), OVDiff (Karazija et al., 2024), and DSG-OV (Wang et al., 2025b). In addition, we include representative methods that combine strong foundation segmentation models, such as SAM-CLIP, to assess the upper performance bound under the unsupervised open-vocabulary setting.

Quantitative Evaluation. Across multiple datasets and evaluation protocols, our method consistently demonstrates strong and robust performance in unsupervised open-vocabulary segmentation. Under the *with-background* setting, PhaseAlign achieves 68.1% mIoU on V21 and 42.1% mIoU on the more challenging C60 benchmark, outperforming all compared methods. For object-level evaluation, it attains 39.1% mIoU, surpassing recent approaches such as CLIP-DINOiser (Wysoczańska et al., 2024) and OVDiff (Karazija et al., 2024), and establishing a new state-of-the-art under this protocol. Under the *without-background* setting, PhaseAlign further reaches 88.0% mIoU on V20 and 41.3% mIoU on C59, remaining competitive with the strongest existing methods. On stuff-dominated benchmarks, our method achieves 29.9% mIoU on the Stuff split, as well as 22.1% and 40.9% mIoU on ADE20K and Cityscapes, respectively, yielding substantial improvements over prior approaches. Overall, these results demonstrate

Table 1. Quantitative Comparison of Recent Open-Vocabulary Segmentation Methods. The frozen setting refers to the visual backbone. Results are reported in mIoU (%). The highest-performing result is highlighted in **bold**, and the second highest in underline. Avg is reported only when all datasets are available; otherwise marked as “–”.

Method	Pub.	Frozen	Visual-Encoder	With background			Without background					Avg.
				V21	C60	Object	V20	C59	Stuff	ADE	City	
ViT-Base for OVS												
CLIP	ICML’21	✗	CLIP	16.4	8.4	5.6	41.9	9.2	4.4	2.9	5.0	11.7
Group ViT	CVPR’22	✗	Custom ViT	50.4	18.7	27.5	79.7	23.4	15.3	9.2	11.1	29.4
TCL	CVPR’23	✗	CLIP	51.2	24.3	30.4	77.5	30.3	19.6	14.9	23.1	33.9
SegCLIP	ICML’23	✓	CLIP	52.6	24.7	26.5	–	–	–	–	–	–
CoDe	CVPR’24	✓	DINO	57.7	30.5	32.3	–	–	23.9	17.7	28.9	–
SAM-CLIP	ECCV’24	✓	SAM	60.6	29.2	–	–	–	–	17.1	–	–
CLIP-DINOiser	ECCV’24	✓	CLIP	62.1	32.4	34.8	80.9	35.9	24.6	20.0	31.7	40.3
OVDiff	ECCV’24	✓	CLIP	66.3	29.7	34.6	80.9	32.9	20.3	14.1	23.4	37.8
LaVG	ECCV’24	✓	DINO	62.1	31.6	34.2	82.5	34.7	23.2	15.8	26.2	38.8
PixelCLIP	NeurIPS’24	✓	CLIP	–	–	–	85.9	37.9	23.6	18.7	27.2	–
ProxyCLIP	ECCV’24	✓	CLIP+Dino	61.3	35.3	<u>37.5</u>	80.3	39.1	<u>26.5</u>	20.2	<u>38.1</u>	42.3
CLIPSurgery	PR’25	✓	CLIP	–	29.3	–	–	–	21.9	–	31.4	–
DeCLIP	CVPR’25	✓	CLIP	59.7	35.3	36.4	85.0	39.2	25.3	<u>21.9</u>	32.8	41.2
DSG-OV	CVPR’25	✓	SAM	<u>67.7</u>	<u>40.8</u>	36.3	85.0	46.5	25.7	20.7	36.0	<u>44.1</u>
PhaseAlign	ICML’26	✓	DINOv2(reg)	68.1	42.1	39.1	88.0	<u>41.3</u>	29.9	22.1	40.9	46.4
ViT-Large for OVS												
MaskCLIP	ECCV’22	✓	CLIP	23.3	11.7	7.2	29.4	12.4	8.8	7.2	11.5	13.9
FreeDA	CVPR’24	✓	DINOv2	45.3	34.3	26.7	75.2	<u>39.0</u>	<u>27.0</u>	<u>21.3</u>	33.1	41.4
SCLIP	ECCV’24	✓	CLIP	47.8	23.8	26.9	76.3	27.4	18.7	11.8	23.9	–
ProxyCLIP	ECCV’24	✓	CLIP+DINOv2(reg)	59.4	34.6	<u>39.0</u>	<u>85.8</u>	37.6	25.6	22.5	<u>37.5</u>	<u>44.5</u>
SCLIP + FreeCP	ICCV’25	✓	CLIP	<u>65.8</u>	<u>35.3</u>	37.2	84.3	38.0	24.9	18.4	33.3	42.1
PhaseAlign	ICML’26	✓	DINOv2(reg)	67.9	41.4	39.3	88.9	41.6	30.3	22.5	41.2	46.6

strong generalization across datasets with diverse category distributions, with particularly pronounced advantages on complex scenes dominated by stuff classes.

Qualitative Evaluation. Figure 4 compares PhaseAlign with FreeDA (Barsellotti et al., 2024) and DSG-OV (Wang et al., 2025b) on VOC, ADE20K, COCO-Stuff, and in-the-wild web images. FreeDA often exhibits noisy activations and blurred boundaries, with over-diffusion in complex scenes. DSG-OV produces accurate object localization but frequently suffers from incomplete structures and spurious responses, especially in fine-grained regions and background suppression. In contrast, PhaseAlign consistently yields more coherent regions, sharper boundaries, and more complete object shapes, demonstrating improved robustness on stuff categories and cross-domain web images without additional supervision.

4.4. Ablation Studies

Scale Sensitivity Test. To evaluate the robustness of different cross-modal similarity formulations to feature magnitude perturbations, we apply multiplicative scaling to visual token embeddings during inference, $v' = \alpha v$, $\alpha \in$

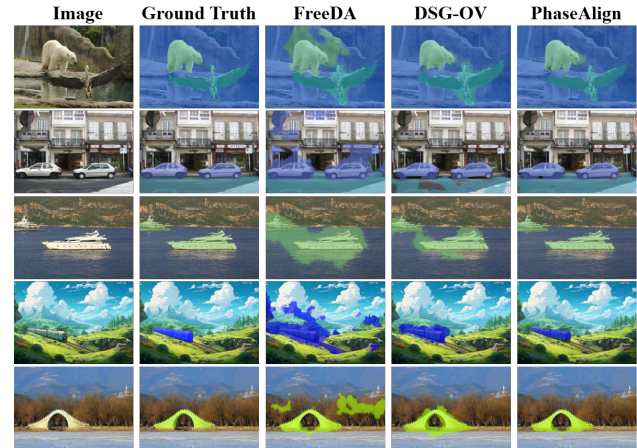


Figure 4. Qualitative comparison between PhaseAlign and existing methods, including FreeDA and DSG-OV, on PASCAL VOC, ADE20K, COCO-Stuff, and in-the-wild datasets.

$\{0.5, 1.0, 2.0, 4.0\}$, while keeping all other components fixed. We compare cosine similarity, inner product similarity, phase-only alignment, and the proposed hybrid similarity S_f . Performance is measured using mIoU and Boundary F-score (BFScore). As shown in Fig. 5, magnitude-dependent similarities exhibit pronounced performance fluc-

tuations under feature scaling, particularly in boundary quality, whereas phase-based and hybrid formulations remain stable across all scaling factors. These results validate the scale robustness of complex phase alignment for pixel-level cross-modal matching.

Table 2. Ablation on Cross-modal Similarity Paradigms. Performance is reported in terms of mIoU (%) and BFScore.

Similarity Formulation	mIoU $\uparrow$	BFScore $\uparrow$
Cosine-only	30.1	7.46
Inner Product-only	26.1	3.55
Phase-only (S_{phase})	37.3	8.55
Hybrid (S_f)	40.9	8.65

Table 3. Ablation Study on Individual Components. Performance is reported in terms of mIoU (%).

+CPA	+AAM	+SACP	ADE	City	Stuff	Avg.
$\times$	$\times$	$\times$	14.3	28.4	22.8	21.8
$\times$	$\checkmark$	$\times$	17.2	33.5	23.5	24.7
$\times$	$\times$	$\checkmark$	18.1	34.1	25.2	25.8
$\checkmark$	$\times$	$\times$	16.5	35.4	24.3	25.4
$\checkmark$	$\checkmark$	$\times$	18.2	38.2	26.1	27.5
$\checkmark$	$\times$	$\checkmark$	19.4	37.3	27.3	28.0
$\checkmark$	$\checkmark$	$\checkmark$	22.1	40.9	29.9	31.0

Table 4. Ablation on Projection Design. Performance is reported in terms of mIoU (%).

Projection	ADE	City	Stuff	Avg.
MLP	16.5	35.4	24.3	25.4
MLP+Conv	18.2	38.2	26.1	27.5
SACP	22.1	40.9	29.9	31.0

Ablation of Alignment Paradigms. To verify that the gains of PhaseAlign are not merely due to similarity re-weighting, we conduct a controlled ablation where we keep the network architecture, training schedule, and all hyper-parameters fixed, and only replace the cross-modal similarity formulation used to generate pixel-wise matching scores. We compare four representative paradigms: cosine similarity (Cosine-only), inner product similarity (Inner Product-only), phase-only alignment (S_{phase}), and our final hybrid similarity (S_f). We report both mIoU and Boundary F-score (BFScore) to reflect region overlap and boundary alignment quality, respectively. As shown in Fig. 5, phase-only alignment already yields a substantial improvement over magnitude-dependent baselines, while the hybrid formulation further improves performance without sacrificing stability, indicating that phase-based alignment provides an indispensable supervision signal for pixel-level cross-modal matching. Table 2 compares different similarity formulations under identical training settings. Magnitude-dependent similarities perform significantly worse, especially in boundary quality, whereas phase-only alignment yields clear improvements. The hybrid formulation further enhances performance, con-

Table 5. Controlled comparison under matched backbone, training data, and prior settings. All methods use the same DINOv2-reg visual backbone, the same training data, and no additional external priors. PhaseAlign consistently achieves the best performance, indicating the effectiveness of the proposed phase-based alignment mechanism.

Method	Visual Backbone	ADE	City	Stuff
FreeDA	DINOv2-reg	21.0	38.6	27.1
DeCLIP	DINOv2-reg	20.3	33.4	25.6
PhaseAlign	DINOv2-reg	22.1	40.9	29.9

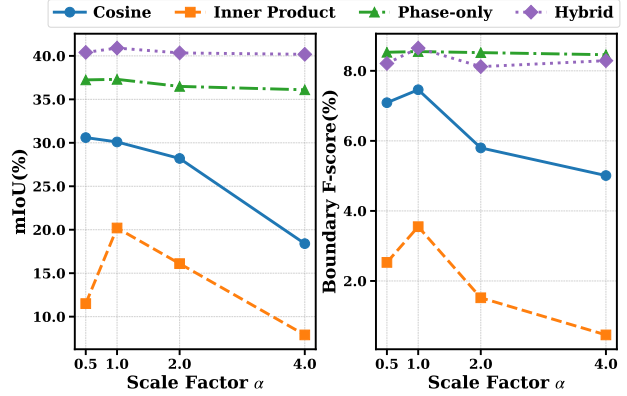


Figure 5. Scale sensitivity under visual feature scaling $v' = \alpha v$. Phase-based and hybrid similarities remain stable across scaling factors, while magnitude-dependent similarities exhibit pronounced performance fluctuations.

firming the essential role of phase-based alignment in pixel-level cross-modal matching.

Effect of Attention-guided Affinity Modeling. As shown in Table 3, introducing the *Attention-guided Affinity Modeling* (AAM) module consistently improves segmentation performance over the CPA baseline. Specifically, enabling AAM alone increases mIoU from 16.5 / 35.4 / 24.3 to 18.2 / 38.2 / 26.1 on ADE / City / Stuff, respectively. This improvement indicates that affinity diffusion guided by ViT attention effectively enhances boundary consistency and long-range structural connectivity. By propagating high-confidence activations along semantically coherent regions, AAM alleviates fragmented predictions and refines region completeness, validating its role as a structure-aware refinement module rather than a representation-level alignment mechanism.

Effect of Spatial-Aware Cross-modal Projection. Table 3 further shows that enabling only the *Spatial-Aware Cross-modal Projection* (SACP) module improves mIoU to 19.4 / 37.3 / 27.3 on ADE / City / Stuff, with more pronounced gains on ADE and Stuff categories, highlighting the importance of spatial-aware alignment in complex scenes. To exclude the possibility that such gains merely arise from a stronger projection head, we conduct a parameter-matched

ablation on projection design in Table 4. Replacing SACP with standard MLP or MLP+Conv projections yields only limited improvements, while SACP achieves substantially higher performance across all datasets. This confirms that the effectiveness of SACP stems from its explicit modeling of local spatial contrast and neighborhood consistency, rather than increased model capacity, thereby supporting its role in improving cross-modal alignment quality.

Controlled comparison under matched settings. To further examine whether the performance gain comes from the proposed alignment mechanism rather than differences in backbone, training data, or additional priors, we conduct a controlled comparison under strictly matched settings. As shown in Table 5, all methods adopt the same pretrained visual backbone, use the same training data, and do not introduce any additional external priors. Under this fair setting, PhaseAlign consistently outperforms FreeDA and DeCLIP on ADE, Cityscapes, and COCO-Stuff. Specifically, PhaseAlign achieves 22.1, 40.9, and 29.9 mIoU on the three benchmarks, respectively, surpassing FreeDA by 1.1, 2.3, and 2.8 points. These results demonstrate that PhaseAlign remains competitive under controlled conditions, supporting that the observed gains are not simply due to differences in backbone choice, training data, or external priors.

5. Limitations and Future Work

PhaseAlign is currently evaluated mainly on 2D open-vocabulary semantic segmentation and still relies on pretrained visual-textual representations, while introducing moderate computational overhead. Future work will extend it to 3D open-vocabulary scene understanding by exploring phase-based alignment for spatial structure, multi-view consistency, geometry-language correspondence, and more challenging domain-shifted scenarios involving point clouds and neural scene representations.

6. Conclusion

This work revisits cross-modal alignment in open-vocabulary semantic segmentation and identifies magnitude-sensitive similarity as a key factor limiting fine-grained prediction. To address this issue, we propose PhaseAlign, which formulates visual-textual alignment in the complex domain and measures semantic consistency via phase differences, effectively decoupling alignment from embedding magnitude. Building on this formulation, we incorporate spatial-aware cross-modal projection and attention-guided affinity modeling to capture local structure and boundary coherence. Extensive experiments across multiple benchmarks demonstrate that PhaseAlign achieves state-of-the-art performance, improving both alignment stability and pixel-level structural fidelity.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Barsellotti, L., Amoroso, R., Cornia, M., Baraldi, L., and Cucchiara, R. Training-free open-vocabulary segmentation with offline diffusion-augmented prototype generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3689–3698, 2024.
- Born, M. and Wolf, E. *Principles of Optics*. Cambridge University Press, Cambridge, UK, 7th edition, 1999.
- Bucher, M., Vu, T.-H., Cord, M., and Pérez, P. Zero-shot semantic segmentation. *Advances in Neural Information Processing Systems*, 32, 2019.
- Caesar, H., Uijlings, J., and Ferrari, V. Coco-stuff: Thing and stuff classes in context. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1209–1218, 2018.
- Cha, J., Mun, J., and Roh, B. Learning to generate text-grounded mask for open-world semantic segmentation from only image-text pairs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11165–11174, 2023.
- Cheng, B., Misra, I., Schwing, A. G., Kirillov, A., and Girdhar, R. Masked-attention mask transformer for universal image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 1290–1299, 2022.
- Cherti, M., Beaumont, R., Wightman, R., Wortsman, M., Ilharco, G., Gordon, C., Schuhmann, C., Schmidt, L., and Jitsev, J. Reproducible scaling laws for contrastive language-image learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 2818–2829, 2023.
- Cho, S., Shin, H., Hong, S., Arnab, A., Seo, P. H., and Kim, S. Cat-seg: Cost aggregation for open-vocabulary semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4113–4123, 2024.
- Cordts, M., Omran, M., Ramos, S., Rehfeld, T., Enzweiler, M., Benenson, R., Franke, U., Roth, S., and Schiele, B. The cityscapes dataset for semantic urban scene understanding. In *Proceedings of the IEEE conference on*

- computer vision and pattern recognition*, pp. 3213–3223, 2016.
- Darcet, T., Oquab, M., Mairal, J., and Bojanowski, P. Vision transformers need registers. In Kim, B., Yue, Y., Chaudhuri, S., Fragkiadaki, K., Khan, M., and Sun, Y. (eds.), *International Conference on Learning Representations*, volume 2024, pp. 2632–2652, 2024.
- Dong, X., Bao, J., Zheng, Y., Zhang, T., Chen, D., Yang, H., Zeng, M., Zhang, W., Yuan, L., Chen, D., et al. Maskclip: Masked self-distillation advances contrastive language-image pretraining. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10995–11005, 2023.
- Dosovitskiy, A. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- Everingham, M., Van Gool, L., Williams, C. K. I., Winn, J., and Zisserman, A. The PASCAL Visual Object Classes Challenge 2012 (VOC2012) Results. <http://www.pascal-network.org/challenges/VOC/voc2012/workshop/index.html>, 2012.
- Fu, Z., Zhang, L., Xia, H., and Mao, Z. Linguistic-aware patch slimming framework for fine-grained cross-modal alignment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26307–26316, 2024.
- Han, J., Zhao, X., Li, L., Wang, C., and Li, S. Adaptive prototype learning for weakly supervised semantic segmentation. In *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*, pp. 215–229. Springer, 2025.
- Huang, J., Wang, J., Sun, Q., and Zhang, H. Attention-based class activation diffusion for weakly-supervised semantic segmentation. *arXiv preprint arXiv:2211.10931*, 2022.
- Jia, C., Yang, Y., Xia, Y., Chen, Y.-T., Parekh, Z., Pham, H., Le, Q., Sung, Y.-H., Li, Z., and Duerig, T. Scaling up visual and vision-language representation learning with noisy text supervision. In *International conference on machine learning*, pp. 4904–4916. PMLR, 2021.
- Jing, D., He, X., Luo, Y., Fei, N., Wei, W., Zhao, H., Lu, Z., et al. Fineclip: Self-distilled region-based clip for better fine-grained understanding. *Advances in Neural Information Processing Systems*, 37:27896–27918, 2024.
- Jose, C., Moutakanni, T., Kang, D., Baldassarre, F., Darcet, T., Xu, H., Li, D., Szafraniec, M., Ramamonjisoa, M., Oquab, M., et al. Dinov2 meets text: A unified framework for image-and pixel-level vision-language alignment. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 24905–24916, 2025.
- Kang, R., Song, Y., Gkioxari, G., and Perona, P. Is clip ideal? no. can we fix it? yes! *arXiv preprint arXiv:2503.08723*, 2025.
- Karazija, L., Laina, I., Vedaldi, A., and Rupprecht, C. Diffusion models for open-vocabulary segmentation. In *European Conference on Computer Vision*, pp. 299–317. Springer, 2024.
- Lan, M., Chen, C., Ke, Y., Wang, X., Feng, L., and Zhang, W. Proxyclick: Proxy attention improves clip for open-vocabulary segmentation. In *European Conference on Computer Vision*, pp. 70–88. Springer, 2024.
- Liu, Y., Ge, P., Liu, Q., and Huang, D. Multi-grained cross-modal alignment for learning open-vocabulary semantic segmentation from text supervision. *arXiv preprint arXiv:2403.03707*, 2024.
- Liu, Y., Ge, P., Wang, G., Liu, Q., and Huang, D. Multi-grained contrastive learning for text-supervised open-vocabulary semantic segmentation. *ACM Transactions on Multimedia Computing, Communications and Applications*, 21(3):1–21, 2025.
- Mottaghi, R., Chen, X., Liu, X., Cho, N.-G., Lee, S.-W., Fidler, S., Urtasun, R., and Yuille, A. The role of context for object detection and semantic segmentation in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 891–898, 2014.
- Mukhoti, J., Lin, T.-Y., Poursaeed, O., Wang, R., Shah, A., Torr, P. H., and Lim, S.-N. Open vocabulary semantic segmentation with patch aligned contrastive learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 19413–19423, 2023.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.
- Rao, Y., Zhao, W., Chen, G., Tang, Y., Zhu, Z., Huang, G., Zhou, J., and Lu, J. Denseclip: Language-guided dense prediction with context-aware prompting. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 18082–18091, 2022.
- Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pp. 234–241. Springer, 2015.
- Wang, F., Mei, J., and Yuille, A. Sclip: Rethinking self-attention for dense vision-language inference. In *European Conference on Computer Vision*, pp. 315–332. Springer, 2024.

- Wang, J., Li, X., Zhang, J., Xu, Q., Zhou, Q., Yu, Q., Sheng, L., and Xu, D. Diffusion model is secretly a training-free open vocabulary semantic segmenter. *IEEE Transactions on Image Processing*, 2025a.
- Wang, M., Ding, H., Liew, J. H., Liu, J., Zhao, Y., and Wei, Y. Segrefiner: Towards model-agnostic segmentation refinement with discrete diffusion process. *arXiv preprint arXiv:2312.12425*, 2023.
- Wang, Z., Feng, T., Lyu, F., Shang, F., Feng, W., and Wan, L. Dual semantic guidance for open vocabulary semantic segmentation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 20212–20222, 2025b.
- Wu, J.-J., Chang, A. C.-H., Chuang, C.-Y., Chen, C.-P., Liu, Y.-L., Chen, M.-H., Hu, H.-N., Chuang, Y.-Y., and Lin, Y.-Y. Image-text co-decomposition for text-supervised semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 26794–26803, 2024.
- Wysoczańska, M., Siméoni, O., Ramamonjisoa, M., Bursuc, A., Trzcinski, T., and Pérez, P. Clip-dinoiser: Teaching clip a few dino tricks for open-vocabulary semantic segmentation. In *European Conference on Computer Vision*, pp. 320–337. Springer, 2024.
- Xu, J., De Mello, S., Liu, S., Byeon, W., Breuel, T., Kautz, J., and Wang, X. Groupvit: Semantic segmentation emerges from text supervision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 18134–18144, June 2022.
- Xu, J., Liu, S., Vahdat, A., Byeon, W., Wang, X., and De Mello, S. Open-vocabulary panoptic segmentation with text-to-image diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 2955–2966, 2023.
- Yang, J., Luo, K. Z., Li, J., Deng, C., Guibas, L., Krishnan, D., Weinberger, K. Q., Tian, Y., and Wang, Y. Denoising vision transformers. In *European Conference on Computer Vision*, pp. 453–469. Springer, 2024.
- Zhao, H., Puig, X., Zhou, B., Fidler, S., and Torralba, A. Open vocabulary scene parsing. In *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2002–2010, 2017.
- Zhou, B., Zhao, H., Puig, X., Fidler, S., Barriuso, A., and Torralba, A. Scene parsing through ade20k dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 633–641, 2017.

A. Additional Evaluation Metrics

A.1. Boundary F-score (BFScore)

Since PhaseAlign explicitly targets boundary sharpness and structural consistency via phase-based alignment and affinity diffusion, we additionally report boundary-sensitive metrics. Boundary F-score (BFScore) is a standard evaluation metric for assessing boundary quality in semantic segmentation. Unlike region-based metrics such as mIoU, BFScore focuses on the alignment accuracy between predicted and ground-truth object boundaries, making it particularly suitable for evaluating fine-grained structures and contour precision.

Given a predicted segmentation map $\hat{M}$ and the corresponding ground-truth annotation M , boundary maps $\partial\hat{M}$ and ∂M are first extracted using standard edge detection operators. A predicted boundary pixel is considered a true positive if it lies within a tolerance distance δ of any ground-truth boundary pixel. Based on this matching criterion, boundary precision and recall are defined as

$$P_b = \frac{|\partial\hat{M} \cap_{\delta} \partial M|}{|\partial\hat{M}|}, \quad R_b = \frac{|\partial\hat{M} \cap_{\delta} \partial M|}{|\partial M|}, \quad (15)$$

where $\cap_{\delta}$ denotes boundary matching within the tolerance δ .

The BFScore is then computed as the harmonic mean of boundary precision and recall:

$$\text{BFScore} = \frac{2P_b R_b}{P_b + R_b}. \quad (16)$$

Following common practice, the tolerance δ is set proportional to the image resolution to account for minor localization ambiguities along object boundaries.

B. Additional Experiments and Analyses

This appendix provides additional analyses to clarify the design rationale and robustness of PhaseAlign. Rather than introducing auxiliary engineering heuristics, the experiments in this section aim to verify the necessary conditions under which complex phase alignment yields stable and meaningful pixel-level cross-modal correspondence. Specifically, we examine (i) the validity of complex-valued representation design, (ii) the robustness of phase-based similarity composition, and (iii) the role of spatial structure in enabling reliable phase alignment. Together, these analyses demonstrate that the effectiveness of PhaseAlign arises from principled alignment mechanisms, rather than from implementation-specific choices or hyperparameter tuning.

B.1. Validity of Complex Representation Design

We analyze how different real–imaginary decomposition strategies affect semantic continuity and cross-modal alignment stability in the complex domain. Since Complex Phase Alignment (CPA) relies on phase differences to model relative semantic consistency between visual and textual embeddings, the integrity of phase information critically depends on how real and imaginary components are constructed from the original channel representations.

Given visual and textual features $v, t \in \mathbb{R}^C$ projected into a unified embedding space, we consider three representative decomposition schemes: *Half*, *Head–Tail*, and *Interleaved*. The Half and Head–Tail strategies preserve contiguous channel structures when forming the real and imaginary components, whereas the Interleaved scheme disrupts channel locality by alternately assigning channels to the real and imaginary parts. As reported in Table 6, both Half and Head–Tail decompositions achieve identical and consistently strong performance across ADE, Cityscapes, and Stuff benchmarks. In contrast, the Interleaved strategy leads to a substantial degradation in segmentation accuracy. This behavior indicates that *breaking channel continuity destroys phase consistency*, thereby impairing the stability and discriminability of phase-based cross-modal alignment.

These results demonstrate that the effectiveness of CPA does not arise merely from introducing complex-valued representations, but rather from *preserving coherent semantic structure in the real–imaginary decomposition*. Disrupting channel continuity corrupts the underlying phase relationships, confirming that careful complex representation design is a necessary condition for reliable phase-based alignment, rather than an implementation-specific artifact.

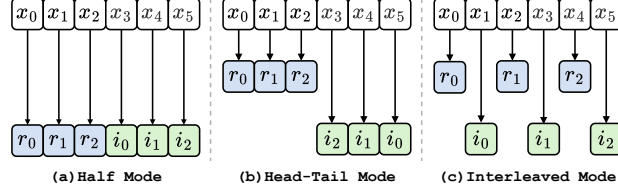


Figure 6. Illustration of different real-imaginary decomposition strategies in the complex-valued space. Half and Head-Tail modes preserve channel continuity when constructing real and imaginary components, whereas the Interleaved mode disrupts local semantic structure by alternating channel assignments, leading to degraded phase consistency.

Table 6. Comparison of Token Representations in Complex-valued Space. Performance is reported in terms of mIoU (%).

Method	ADE	City	Stuff	Avg.
Half	22.1	40.9	29.9	31.0
Head-Tail	22.1	40.9	29.9	31.0
Interleaved	19.6	35.5	26.3	27.1

B.2. Robustness of Phase-based Similarity Composition

We analyze the effect of different similarity weighting strategies to assess whether the performance gains of PhaseAlign rely on careful hyperparameter tuning or reflect an intrinsic robustness of phase-based alignment. Recall that the final cross-modal similarity in CPA is formulated as a weighted combination of phase alignment, global cosine consistency, and in-batch contrastive regularization. A key question is whether the effectiveness of CPA critically depends on a specific weighting configuration.

To this end, we evaluate multiple combinations of similarity weights (w_p, w_c, w_i) under identical training and evaluation settings. The results are reported in Table 7. When only a single similarity component is used, performance consistently degrades across all datasets, indicating that no individual metric alone provides sufficient supervision for stable pixel-level alignment. This observation suggests that complementary regularization terms are necessary to stabilize optimization, but are not the primary drivers of performance.

Notably, configurations that assign dominant or comparable weight to the phase alignment term consistently achieve strong performance, while moderate variations in the auxiliary weights lead to only marginal performance fluctuations. The best overall results are obtained with $(w_p, w_c, w_i) = (1, 1, 1)$; however, nearby configurations exhibit similar behavior, demonstrating that CPA does not rely on a narrowly tuned operating point.

These findings indicate that phase-based alignment governs the optimization dynamics, while cosine similarity and in-batch contrastive terms mainly serve as stabilizing regularizers. The robustness of performance across a broad range of weighting schemes confirms that the advantages of PhaseAlign stem from the intrinsic properties of complex phase alignment, rather than from careful similarity reweighting or hyperparameter selection.

Table 7. Effect of similarity weight combinations in the complex space. Performance is reported in terms of mIoU (%).

No.	(w_p, w_c, w_i)	ADE	City	Stuff	Avg.
0	(0.5, 0.5, 1)	19.2	36.2	28.1	27.8
1	(1, 0, 0)	19.6	37.3	26.4	27.8
2	(0, 1, 0)	20.3	30.6	27.3	26.1
3	(0, 0, 1)	18.2	35.3	28.2	27.2
4	(1, 0, 1)	20.5	38.2	22.1	26.9
5	(1, 1, 1)	22.1	40.9	29.9	31.0

We further observe a similar robustness trend with respect to the temperature parameter τ in phase similarity. As τ increases, performance improves smoothly without abrupt degradation, indicating that phase-based similarity exhibits well-behaved and stable optimization characteristics across different scaling regimes.

B.3. Role of Spatial Structure in Phase Alignment

We further investigate the role of the Spatial-Aware Cross-modal Projection (SACP) under different projection settings to clarify whether its effectiveness stems from increased representational capacity or from improved spatial alignment structure.

Table 8. Sensitivity analysis of the temperature parameter τ in CPA. Performance is reported in terms of mIoU (%).

τ	ADE	City	Stuff	Avg.
0.01	18.5	36.2	25.2	26.6
0.05	19.2	36.7	26.3	27.4
0.10	20.2	39.6	26.4	28.7
0.50	21.2	38.1	27.2	28.8
1.00	22.1	40.9	29.9	31.0

Unlike conventional projection heads designed to enhance feature expressiveness, SACP is introduced to explicitly impose spatial contrast and neighborhood consistency, which are critical for stable phase-based alignment at the pixel level.

To this end, we evaluate three projection configurations: *Text-only*, where SACP is applied exclusively to the textual branch; *Visual-only*, where SACP is applied only to the visual branch; and *Visual & Textual*, where SACP is jointly applied to both modalities. The results are summarized in Table 9.

Across all configurations, incorporating SACP consistently improves segmentation performance, indicating that spatial-aware normalization provides benefits beyond simple feature projection. Applying SACP only to the textual branch yields a clear performance gain, suggesting that spatialized projection helps mitigate alignment bias arising from heterogeneous modality representations. When applied exclusively to the visual branch, SACP leads to more pronounced improvements on structure-heavy benchmarks such as ADE and Cityscapes, highlighting the importance of explicitly modeling local spatial contrast in visual features.

Notably, the best performance is achieved when SACP is applied to both visual and textual branches, substantially outperforming the corresponding baseline without SACP. This result demonstrates that SACP does not merely strengthen individual modality representations, but rather constructs a *spatially compatible alignment space* in which complex phase alignment can operate reliably. By enforcing consistent spatial structure prior to cross-modal matching, SACP enables CPA to leverage phase information more effectively, confirming its role as a structural prerequisite for stable pixel-level cross-modal alignment rather than an auxiliary capacity-enhancing component.

Table 9. Effect of Spatial-Aware Cross-modal Projection (SACP) under different projection settings. Performance is reported in terms of mIoU (%).

Setting	SACP	ADE	City	Stuff	Avg.
Only Textual	✗	18.5	30.1	22.1	23.6
	✓	19.2	32.1	27.1	26.1
Only Visual	✗	18.4	30.3	26.8	25.2
	✓	20.2	34.5	27.2	27.3
Visual & Textual	✗	20.1	36.1	28.3	28.2
	✓	22.1	40.9	29.9	31.0